Circular motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate inward net force to a change in velocity direction.

SUCCESS CRITERIA - I can...

1. Draw tangent velocity and inward acceleration.
2. Use v^2/r or a supplied acceleration.
3. Identify the real interaction providing centripetal force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two balls launch horizontally from the same height at **3** and **15 m/s**. Ignore air resistance. **Compare fall times** and horizontal ranges.

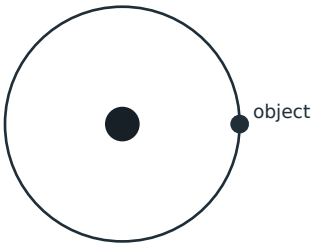
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a_c = v^2/r$; $F_{net} = mv^2/r$.
c means centripetal: toward the center. This is the net inward force, not an extra force.

Worked reasoning: A **2 kg** object moves at **3 m/s** in a circle of radius **3 m**. Inward acceleration is v^2/r=**3 m/s²**; net force is **6 N** inward.

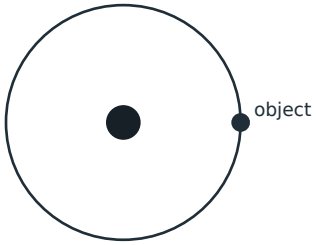
Your turn: Draw tangent velocity and inward net force at two different points.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Add velocity** and acceleration arrows to the marked point on the circle.
2. **Name one interaction** that could provide the inward force.

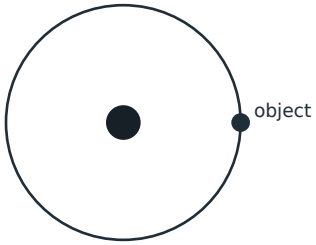


The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

04 Apply the physics

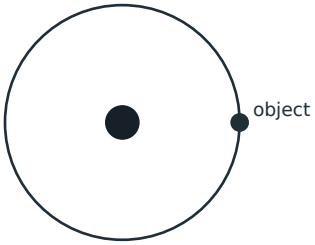
A **2 kg** small classroom car follows a circular track. Its inward acceleration is **3 m/s²**.
1. Draw and **label the inward net force** and **find its magnitude**.
2. How would that force change if inward acceleration doubled?



Marzano check - circle: ① ② ③

05 Explain your reasoning

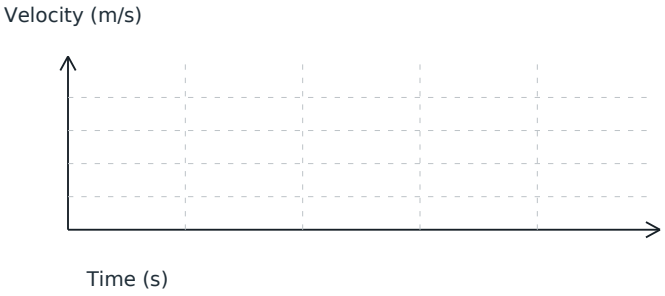
A string breaks while a ball moves in a circle. Neglect other forces over the next instant. **Sketch its initial path after** release.



Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

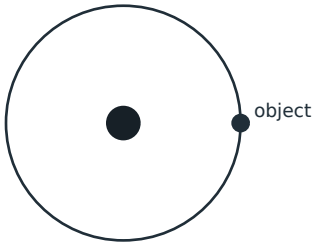
A **8 kg** cart starts from rest under **80 N** net force right for **7 s**. **Find acceleration** and final velocity. **State one thing you** are treating as true.
Label the given and predicted quantities on the diagram.



Marzano check - circle: 1 2 3

07 Exit check

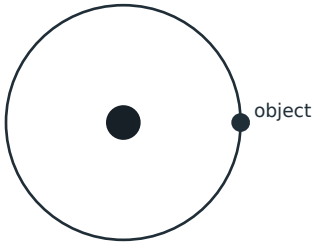
A car turns at steady speed. Can net force be zero?
1. Explain using velocity.
2. **Label the radius** and **show velocity** and inward acceleration on the circle.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**. At the same radius and mass, compare forces for speeds **3** and **6 m/s**. Why is the answer not just twice as much? **Label the radius** and **show velocity** and inward acceleration on the circle.



Marzano check - circle: 1 2 3